SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO4: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Mock Interview **Duration: 1hr**

1. A machine learning engineer applies Reinforcement Learning to solve sequential decision-making problems. Explain how an agent learns optimal behavior through interaction with the environment. **(5 Marks)**

An RL agent learns by **interacting with the environment**:

- It observes the **current state**.
- Selects an **action** using a policy.
- Receives a **reward** and moves to a new state.
- It updates its **value function or policy** based on the reward.
- Repeated interactions help it maximize **cumulative future reward**, leading to optimal behavior.

2. If you are asked to design a Reinforcement Learning solution for a real-time system such as ride-sharing or delivery optimization, explain how you would formulate the problem using a Markov Decision Process. **(5 Marks)**

An MDP can be defined as **(S, A, P, R, γ)**:

- **State (S):** Driver location, available vehicles, traffic, pending requests.

- **Action (A):** Assign driver, choose route, accept/reject request.
- **Transition (P):** New locations and requests after an action.
- **Reward (R):** Profit, completed deliveries, reduced waiting time and fuel.
- **Discount factor (γ):** Controls importance of future rewards.

3. Compare model-free and model-based Reinforcement Learning approaches. Explain their working principles and how they handle uncertainty. **(5 Marks)**

Model-Free	Model-Based
Learns directly from experience	Learns/uses a model of the environment
Does not estimate transitions	Estimates future states and rewards
Requires more interaction	Usually more sample-efficient
Example: Q-Learning, DQN	Example: Dyna-Q, MPC-based RL

4. A recommendation system using Reinforcement Learning is underperforming. Explain how reward sparsity and hyperparameter tuning can be diagnosed and improved. **(5 Marks)**

- **Reward sparsity:** Check whether rewards occur too rarely or provide little learning signal.
- Add **intermediate/shaped rewards** such as clicks, watch time, or user satisfaction.
- Use techniques like **reward normalization** and better exploration.
- Tune **learning rate, discount factor, exploration rate, batch size**, etc.
- Compare performance using validation episodes and learning curves.

5. If you are implementing an RL-based game-playing AI, describe how you would design the learning pipeline and ensure training stability. **(5 Marks)**

Pipeline:
Game Environment → State → Agent → Action → Reward → Replay Buffer → Learning → Updated Policy
For stability:

- Use **experience replay**.
- Use a **target network** for DQN-based methods.
- Apply **reward/gradient clipping**.
- Use suitable learning rates and exploration decay.
- Evaluate periodically using separate test games.

6. A company wants to use Reinforcement Learning for dynamic pricing. Explain how the agent learns from interaction and improves pricing decisions over time. **(5 Marks)**

- **State:** Demand, inventory, competitor prices, time, customer behavior.
- **Action:** Select a product price.
- **Reward:** Revenue/profit while considering demand and customer satisfaction.
- The agent observes the result of each price and updates its policy.
- Through repeated interaction, it learns which prices maximize **long-term profit**.

7. If an RL model is not converging properly, explain how you would identify and fix issues related to learning rate and exploration strategy. **(5 Marks)**

Learning rate diagnosis:

- Too high → unstable or oscillating learning.
- Too low → very slow learning.
- Try a smaller learning rate or use a **learning-rate scheduler**.

Exploration diagnosis:

- Too much exploration → poor exploitation.
- Too little exploration → agent gets stuck in a bad policy.
- Adjust **ε-greedy ε**, usually starting high and gradually decreasing.

8. A robotics system uses Reinforcement Learning for navigation. Explain how the environment and agent interaction leads to optimal path learning. **(5 Marks)**

- **State:** Robot position, sensor readings, obstacles and target location.
- **Actions:** Move forward, backward, left, right, etc.
- **Reward:** Positive for reaching the target; negative for collisions, long paths, or delays.
- The robot repeatedly explores and receives rewards.
- It updates its policy to choose actions that maximize cumulative reward, eventually learning an **optimal/near-optimal path**.

9. If you are asked to improve an RL model with delayed rewards, explain techniques to handle credit assignment problems. **(5 Marks)**

Techniques include:

- **Reward shaping** to provide intermediate feedback.
- **Eligibility traces** to assign rewards to earlier actions.
- **n-step returns** to propagate rewards faster.
- **Temporal-Difference (TD) learning** to update earlier states.
- **Hierarchical RL** to break long tasks into smaller sub-goals.

10. A financial firm uses Reinforcement Learning for trading decisions. Explain how uncertainty and risk can be handled in the learning process. **(5 Marks)**

- Include market volatility, transaction costs and portfolio information in the **state**.
- Use **risk-sensitive rewards** instead of maximizing profit alone.
- Penalize high volatility, large drawdowns and excessive risk.
- Use **Monte Carlo/dropout or ensemble methods** to estimate uncertainty.
- Apply position limits and risk constraints to prevent unsafe trading decisions.

Code of Conduct:

I ***<u>KUSHITHA.G (192425329)</u>*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrate d
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicate s well with minor hesitation	Communication lacks clarity or confidence	Unable to communicat e effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionali sm
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately